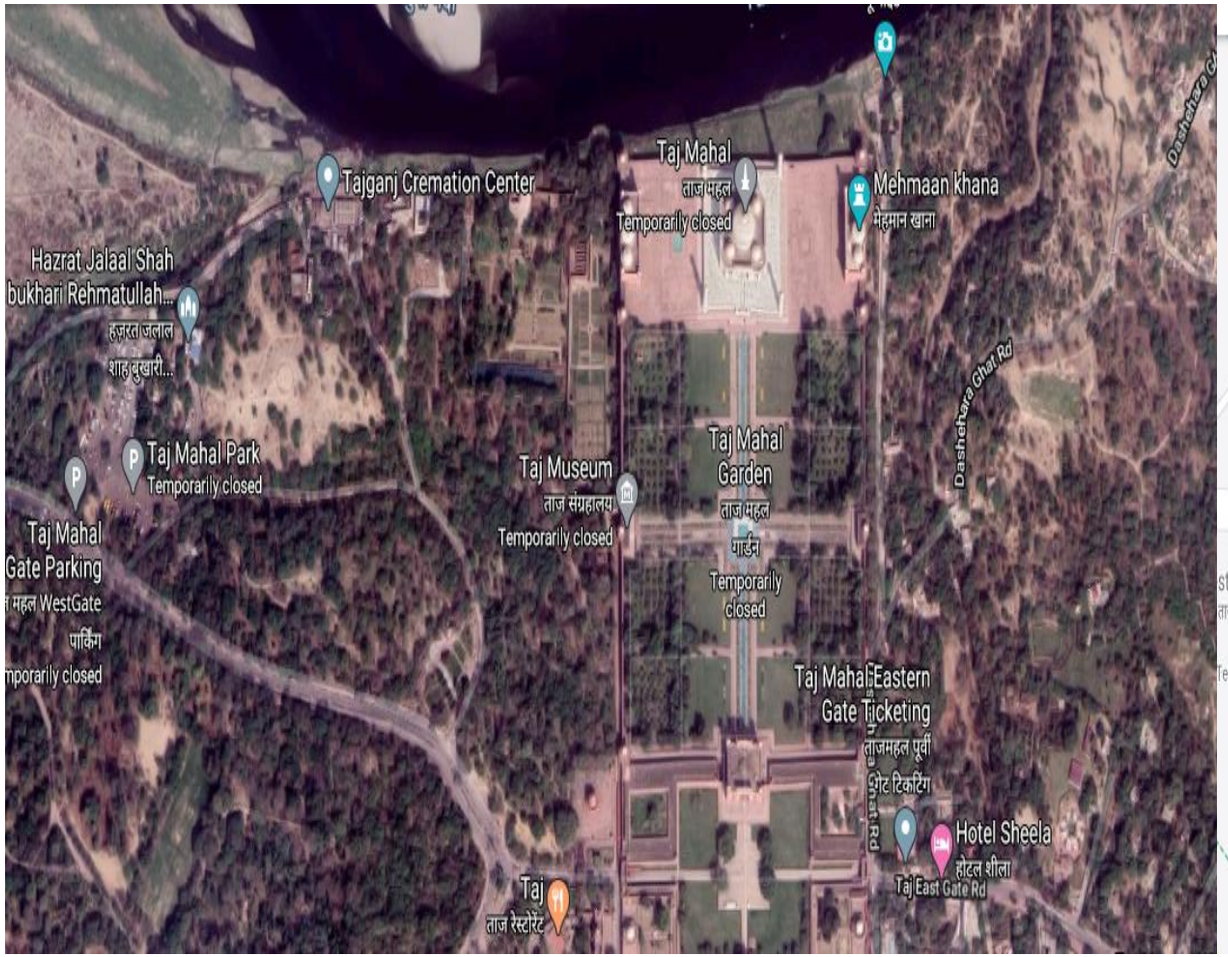
Introduction and Application of Remote Sensing

What is Remote Sensing??



Image Source/Reference/Courtesy: Google

What is Remote Sensing??



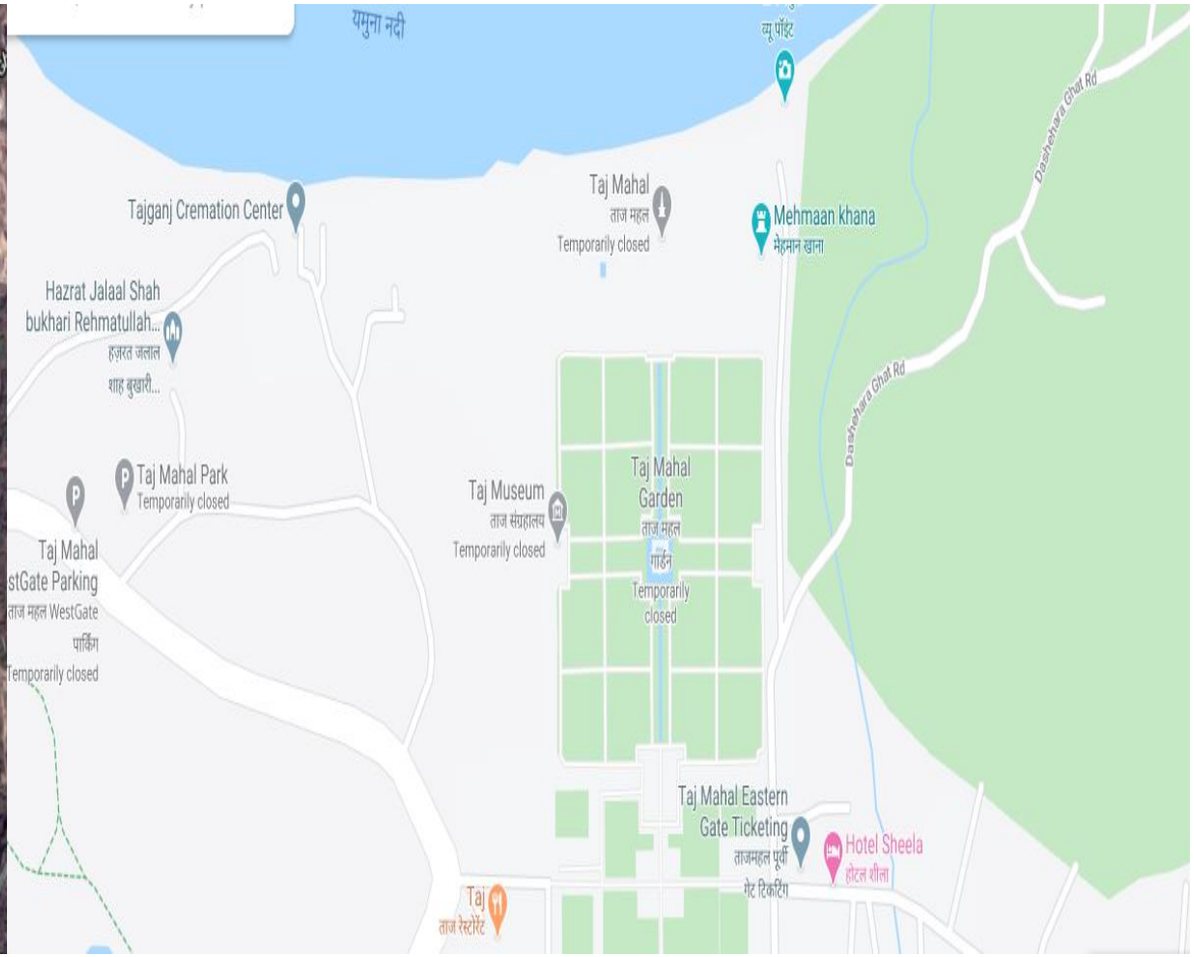


Image Source/Reference/Courtesy: Google

What is Remote Sensing??

- We perceive the surrounding world through our five senses.
- In another word, we are performing Remote Sensing all the time.
- Generally, Remote sensing refers to the activities of recording / observing / perceiving (sensing) objects or events at far away (remote) places.
- In remote sensing, the sensors are not in direct contact with the objects or events being observed.

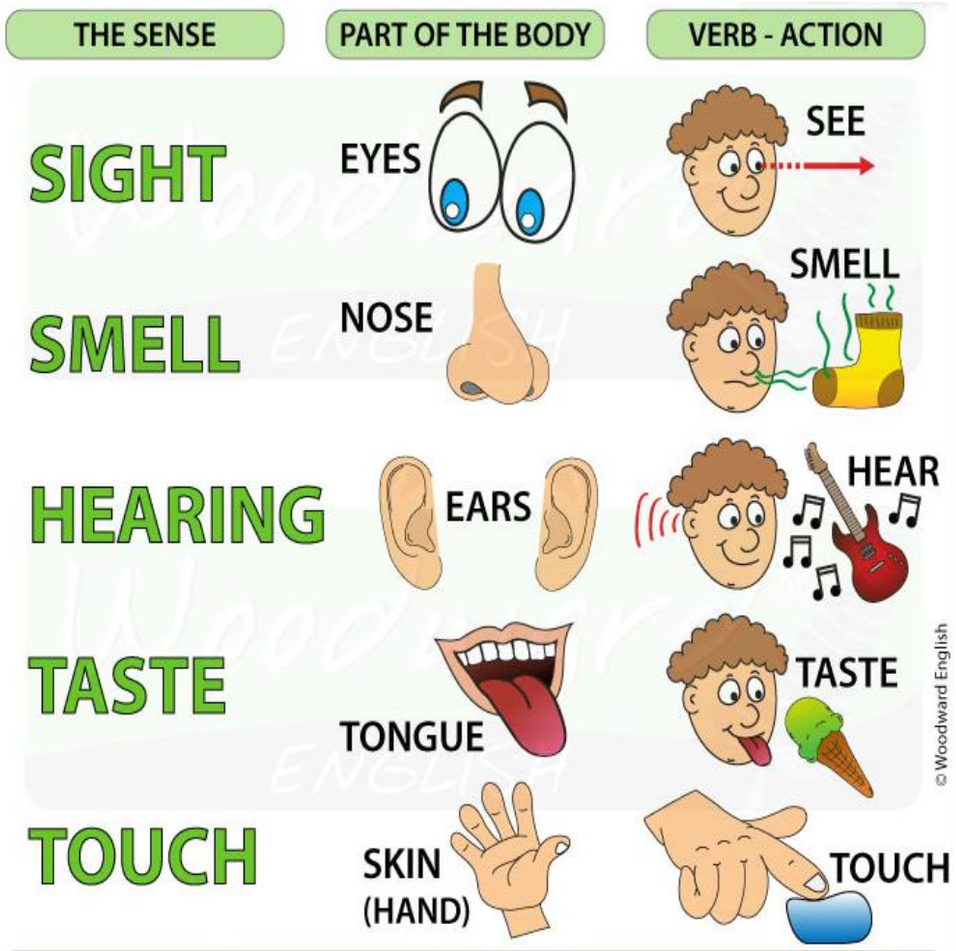


Image Source/Reference/Courtesy: Google

Remote Sensing

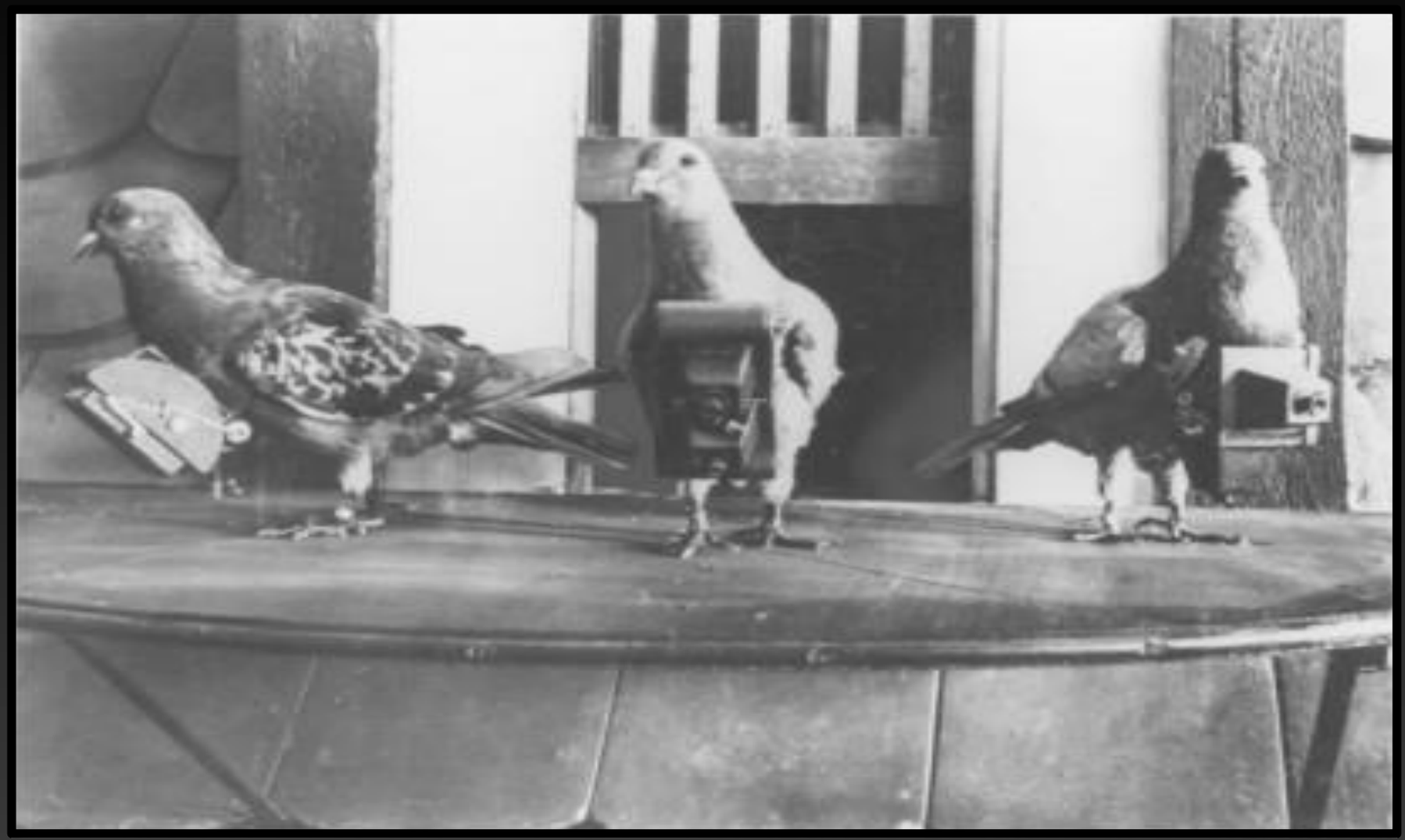
- Remote sensing is the process of acquiring data/information about objects/substances not in direct contact with the sensor.
- It is the collection of information relating to objects without being in physical contact with them.
- An aerial photograph is a common example of a remotely sensed (by camera and film, or now digital) product

Remote Sensing

- Remotely-Sensed data is one of the most important sources of data for GIS.

- RS means - Acquiring data from a distance
- Usually uses electromagnetic energy
  - sunlight, radar, laser
- Originally captured on photographic film
- Recent platforms utilize digital sensors

Early Remote Sensing Platforms



Remote Sensing: How does it work??

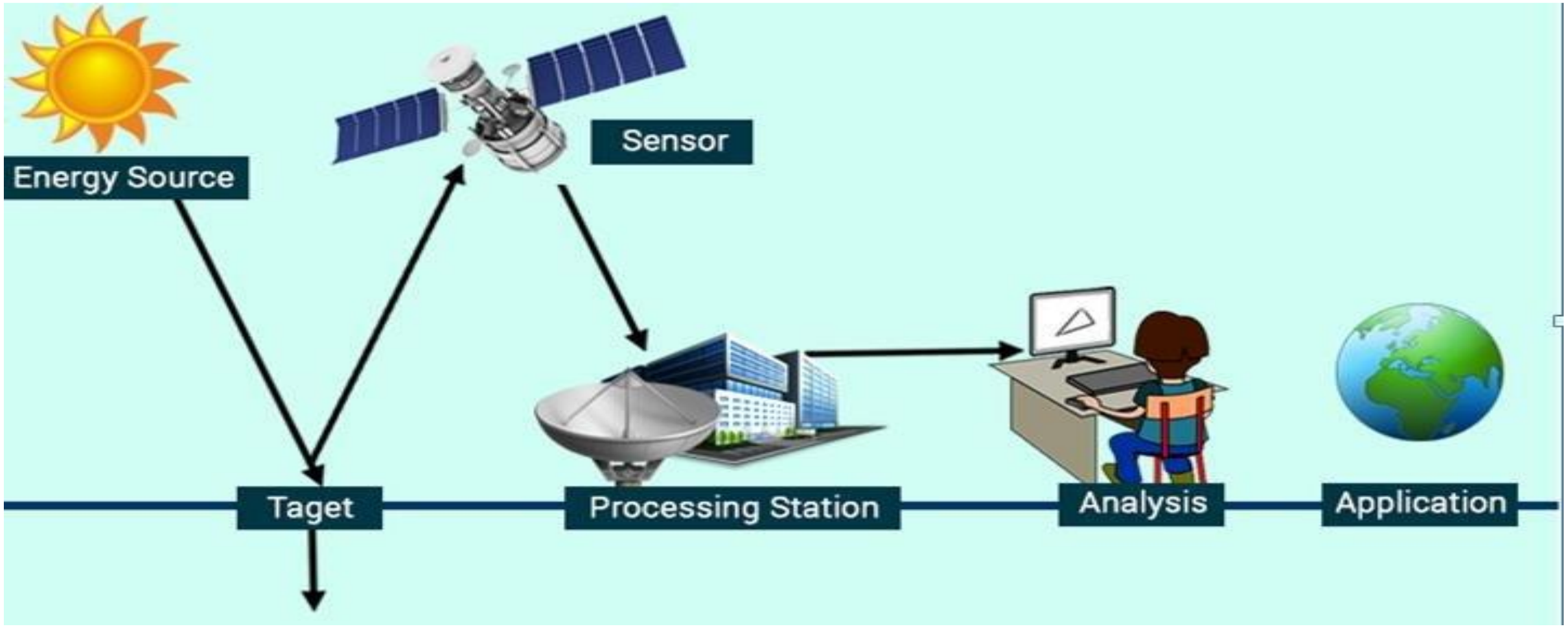


Image Source/Reference/Courtesy: Google

Remote Sensing: How does it work??

- In much of remote sensing, the process involves an interaction between incident radiation and the targets of interest. This can be understood by the use of imaging systems where the following seven elements are involved.
- ***Energy Source or Illumination (A)***
- ***Radiation and the Atmosphere (B)***
- ***Interaction with the Target (C)***
- ***Recording of Energy by the Sensor (D)***
- ***Transmission, Reception and Processing (E)***
- ***Interpretation and Analysis (F)***
- ***Application (G)***

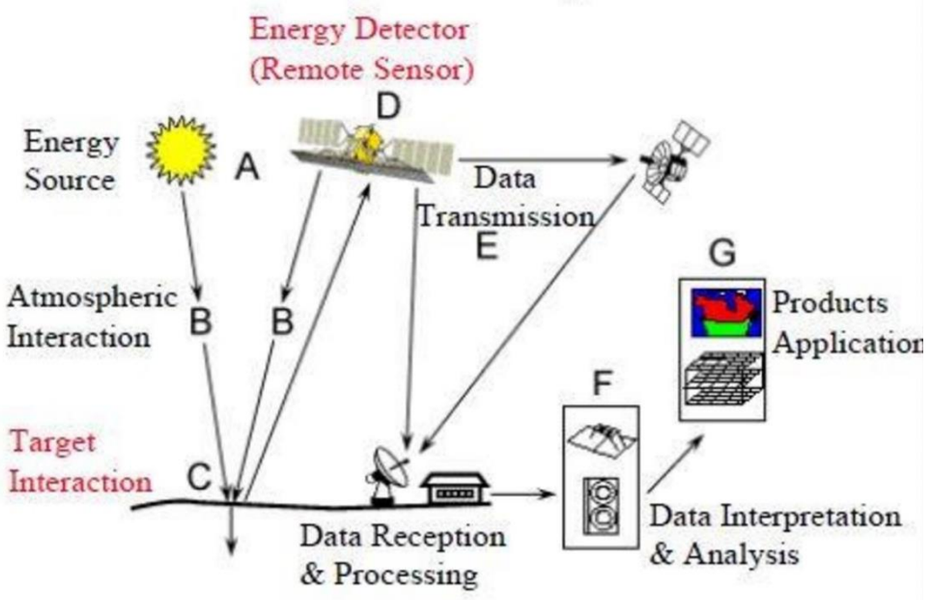


Image Source/Reference/Courtesy: Google

Remote Sensing: How does it work??

- Out of the 3,372 active artificial **satellites** orbiting the Earth as of January 1, 2021, 1,897 belong to the United States.
- This is by far the largest number of any single **country**, **with** their nearest competitor, China, accounting for only 412.
- India has launched 342 satellites for 36 different countries as of 28 February 2021 (49 are **serving various applications for the Country**)

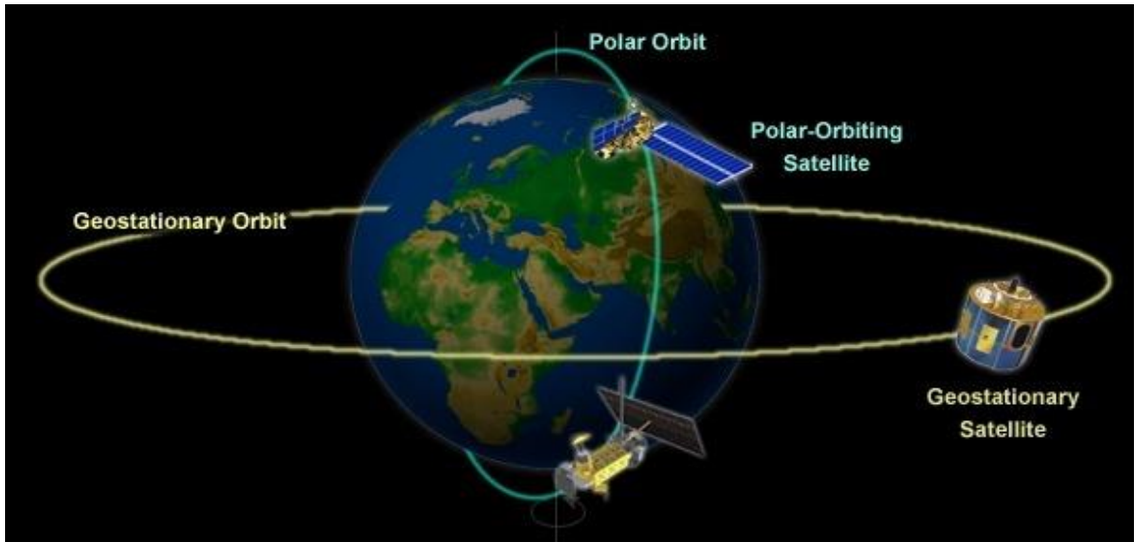
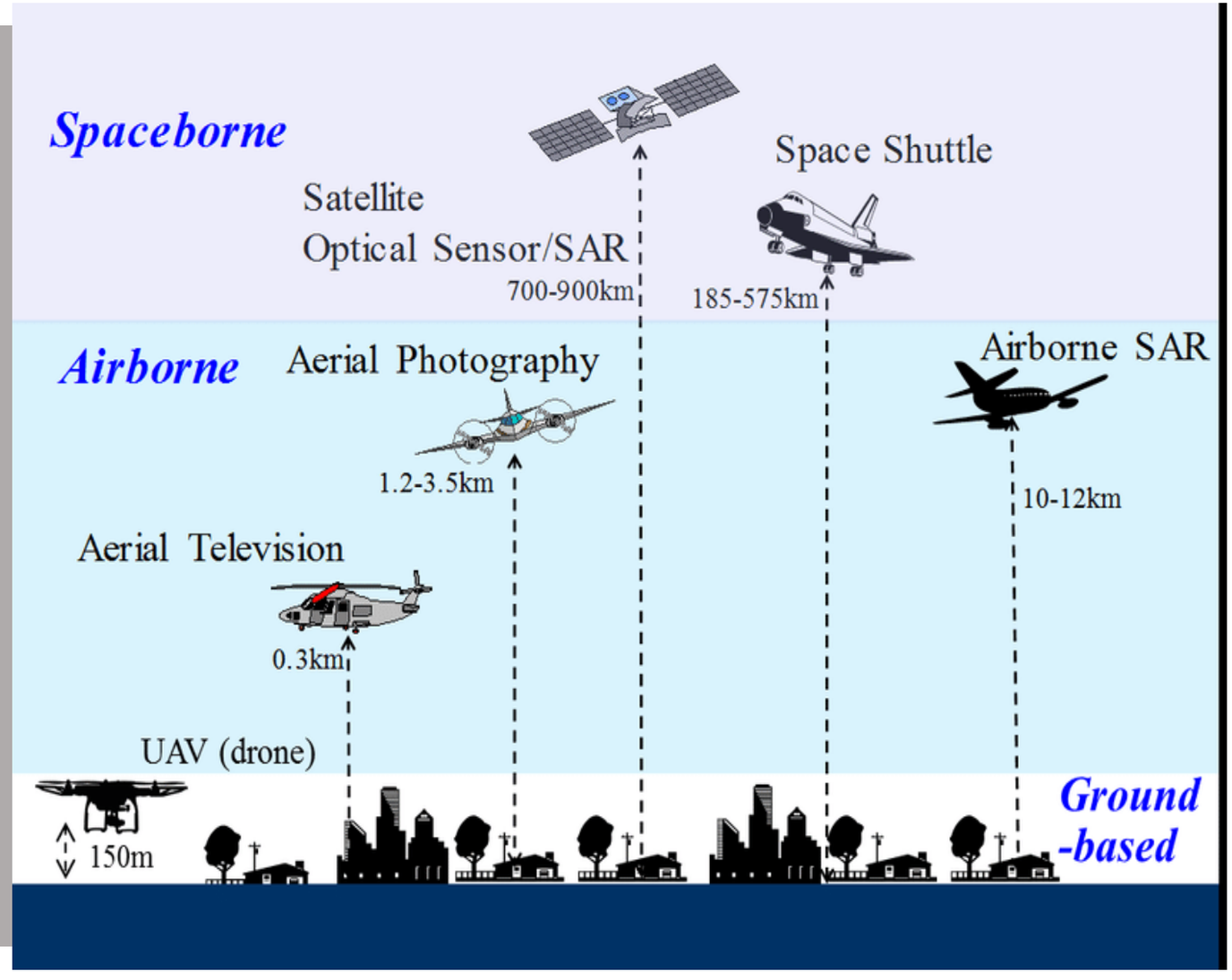




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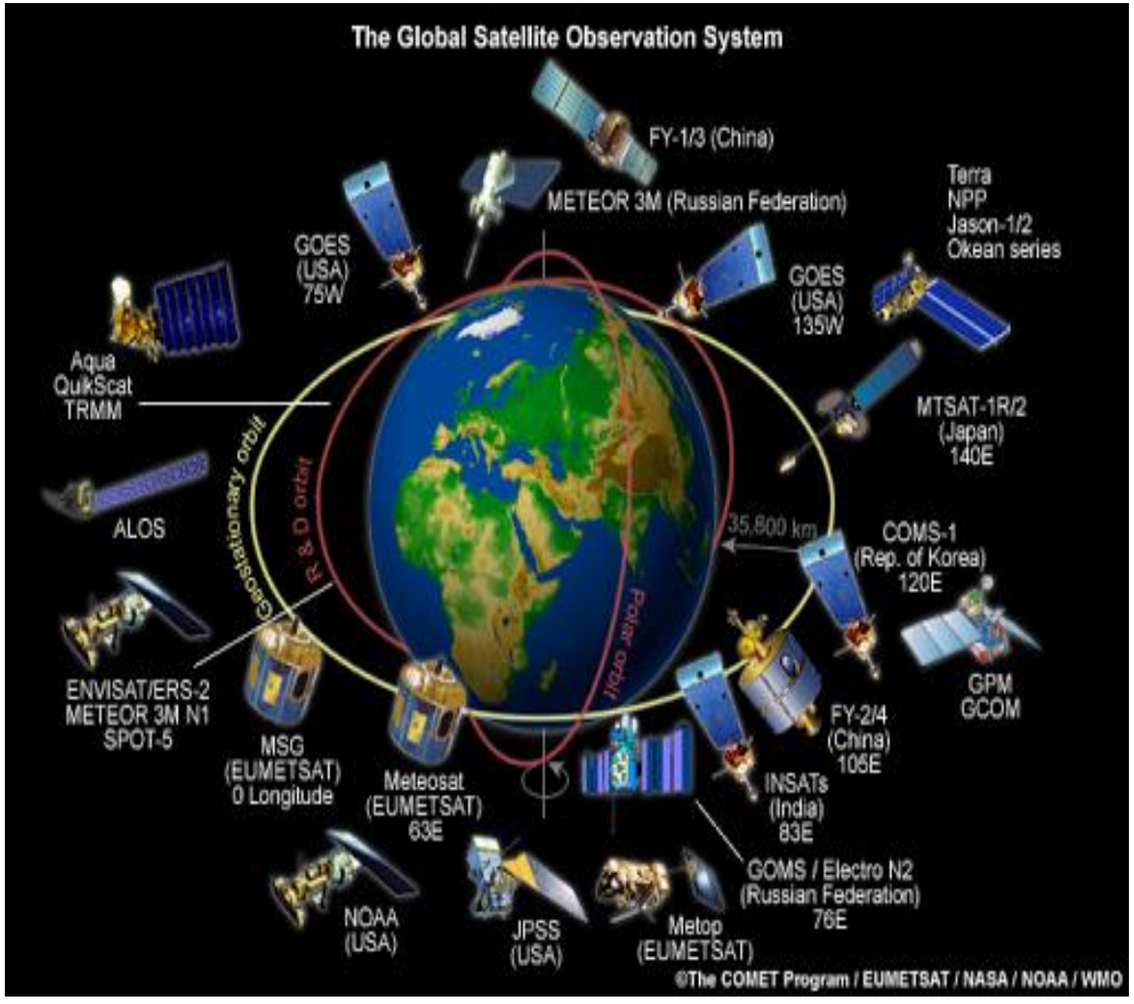
Platforms and Sensors

- For a sensor to collect and record energy reflected or emitted from a target or surface, its must reside on a stable platform away from the target or surface being observed.
- Based on its altitude above earth surface, platforms can be classified as ground-borne, air-borne and space-borne.



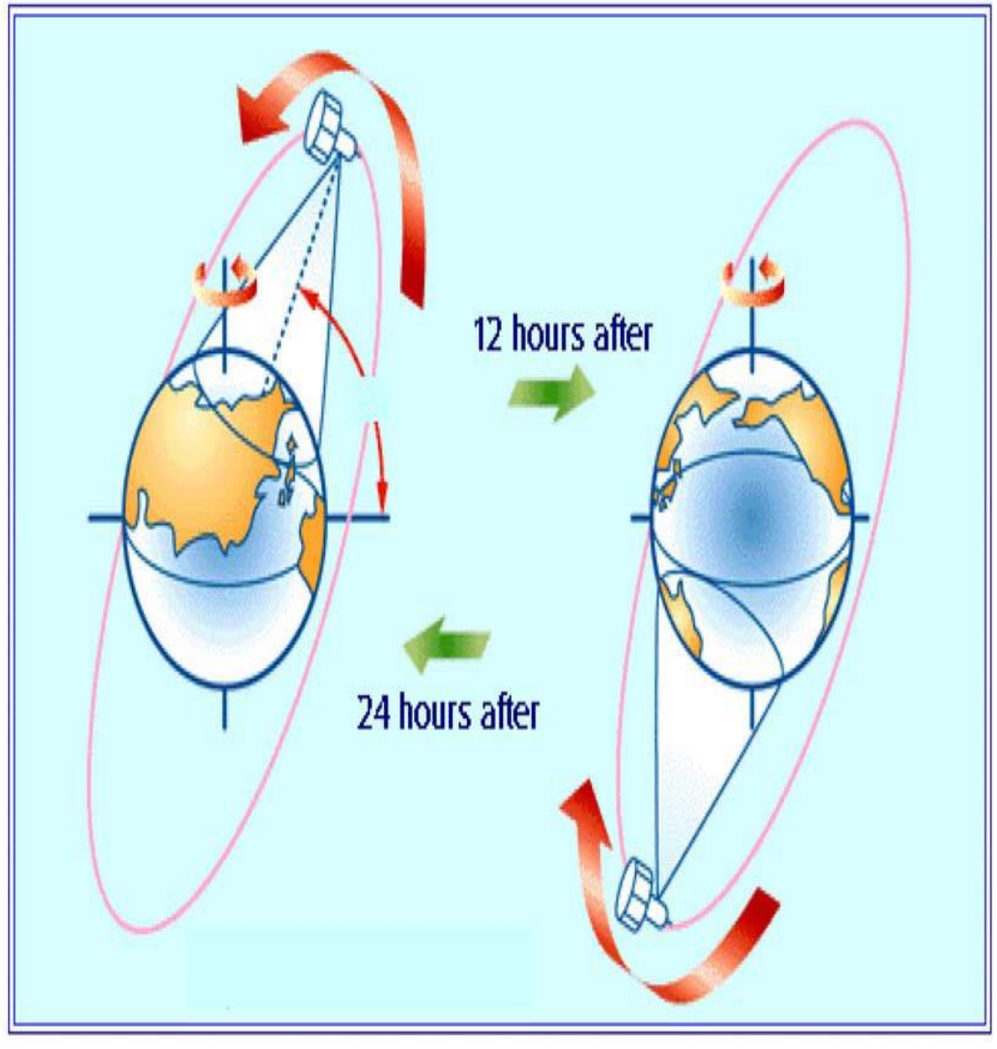
Platforms and Sensors

- **Geostationary satellites:** An equatorial west to east satellite orbiting the earth at an altitude at which it makes one revolution in 24 hours, synchronous with the earth's rotation, hence it gives continuous coverage over the same area day and night. These are mainly used for communication and meteorological applications, for e.g. the INSAT satellites.



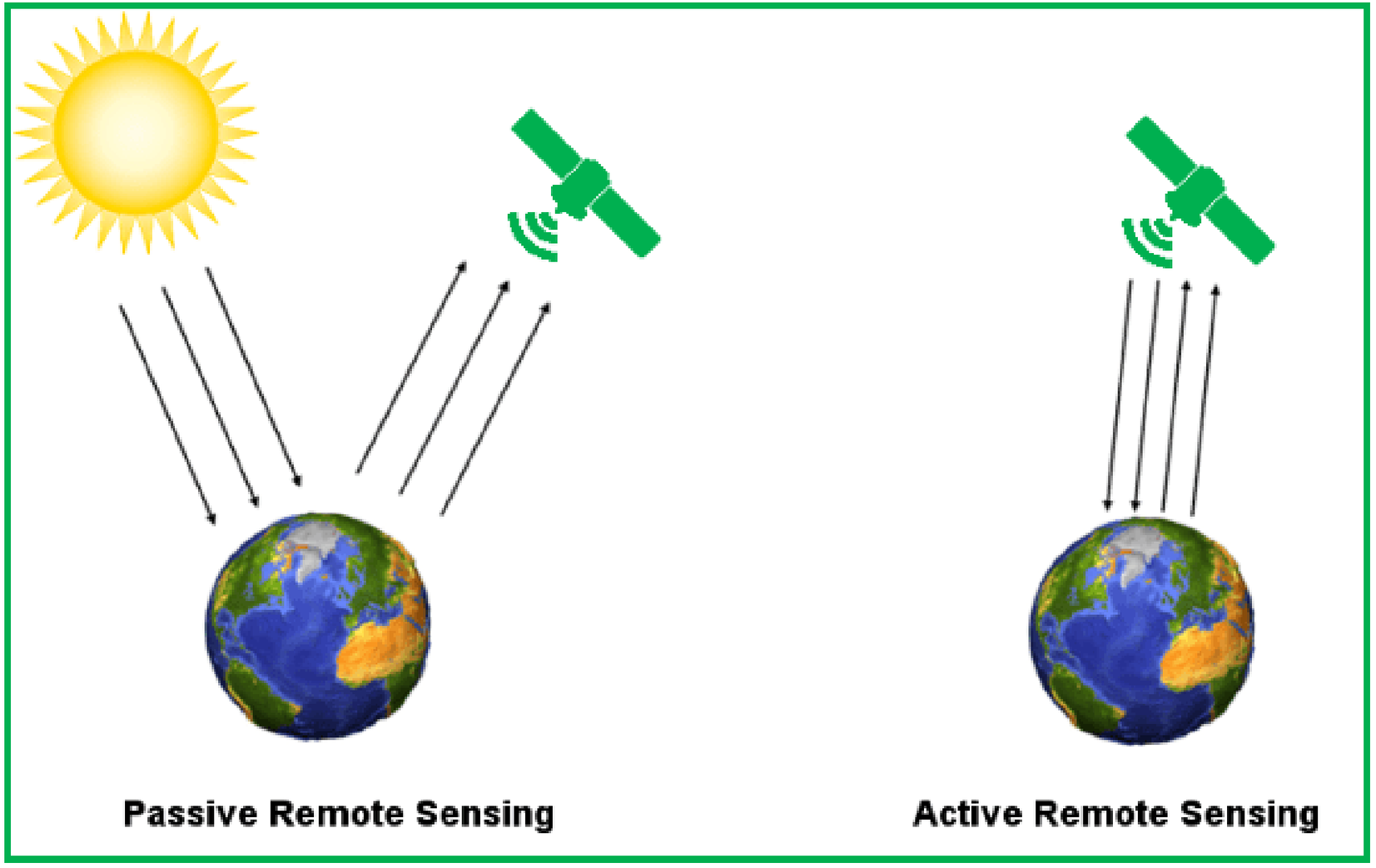
Platforms and Sensors

- **Polar orbiting or Sun-synchronous satellites:** A satellite with inclined north south orbit track westward at a rate such that it covers each area of the world at a constant time of the day called the local sun time as the satellite moves from north to south.
- This ensures the similar illumination conditions while acquiring image over a particular area over a series of days.



Platforms and Sensors

- Sensors used for remote sensing can be broadly classified as those operating in Optical Infrared (OIR) region and those operating in microwave region.
- OIR and microwave sensors can further be subdivided into passive and active depending on the source of energy.
- Active sensors use their own source of energy to illuminate earth surface and a part of it is reflected and received to gather information.
- Passive sensors do not have their own source of energy but instead receive solar electromagnetic energy reflected from surface or energy emitted from surface itself hence it can not be used at night time, except thermal sensors.
- Energy that is naturally emitted (e.g. Thermal energy) can be detected day or night, as long as the amount of energy is large enough to be recorded.


Passive Remote Sensing
Active Remote Sensing

Types of Remote Sensing

- Remote sensing can be broadly classified into three types with respect to the wavelength region and type of sensor involved for data acquisition; viz. Optical (Visible and Reflective Infrared), Thermal Infrared and Microwave.

Optical Remote Sensing	Thermal Infrared Remote Sensing	Microwave Remote Sensing
Optical remote sensing involves the use of visible part of the EM spectrum.	The source of radiant energy used in thermal infrared remote sensing is the object itself, because any object with a normal temperature will emit EM radiation with a peak at about 10 µm	In microwave region, there are two types of microwave remote sensing, passive and active.
Energy emitted from sun is used for visible and reflective infrared remote sensing.		In passive microwave remote sensing, the microwave radiation emitted from an object is detected, while in active microwave remote sensing, source of EM radiations and detector are placed on the sensor

Advantages and Disadvantages of Remote Sensing

Advantages	Disadvantages
Provide a continuous, repetitive, large-scale synoptic view relative to traditional point-based field measurements	Remote sensing instruments are expensive to build and operate
Practical way to obtain data from dangerous or inaccessible areas	Remote sensing data are not direct samples of the phenomenon and it must be calibrated against reality. The measurement uncertainty can be large
Relatively cheap and rapid method of acquiring up-to-date information over a large geographical area	Remote sensing data must be corrected geometrically and georeferenced in order to be useful as maps, not only as pictures. This can be easy or complicated
Easy to manipulate with the computer, and combine with other geographic coverage in the GIS.	Remote sensing data interpretation can be difficult, which usually need to understand theoretically how the instruments are making the measurements, need to understand measurement uncertainties, and need to have some knowledge of the phenomena you are sampling.

Who uses Remote Sensing and why?

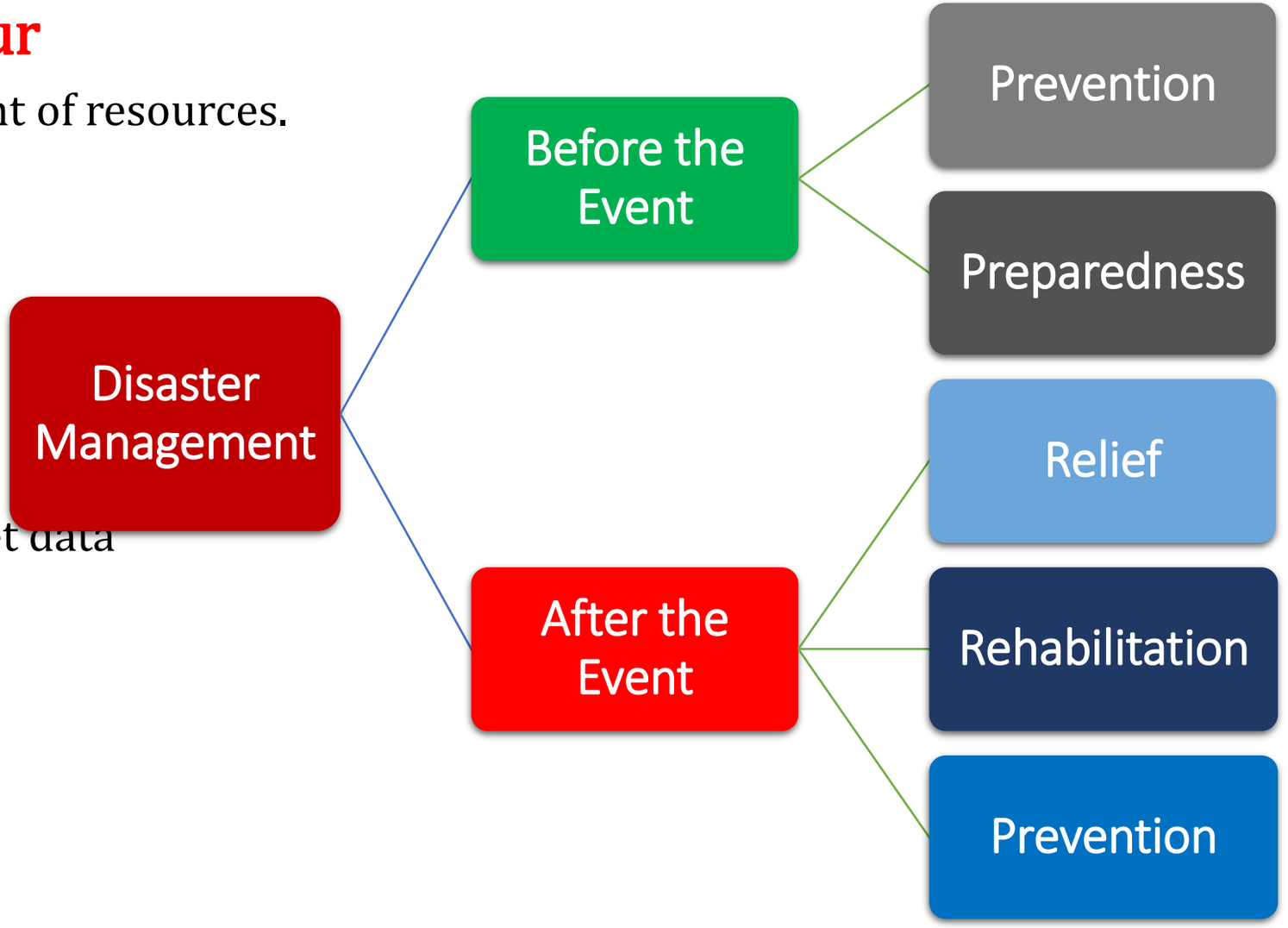
- The geographer, who looks for changes on the Earth's surface that need to be mapped;
- The forester, who needs information about what type of trees are growing and if they have been affected by disease, fire or pollution;
- The environmentalist, who wants to detect, identify and follow the movement of pollutants such as oil slicks on the ocean;
- The geologist, who is interested in finding valuable minerals;
- The farmer, who wants to keep an eye on how his crops are growing and if they've been affected by drought, floods, disease or pests

Who uses Remote Sensing and why?

- The ship captain, who needs to find the best route through the northern ice packs;
- The firefighter, who sends out his crews based on information about the size and movement of a forest fire.
- And there are many more ways to use remote sensing

What we are looking for??

- **Demand/Need of an Hour**
  - ✓ Monitoring and management of resources.
  - ✓ Sustainable development.
  - ✓ Disaster mitigation
- **Requirement:**
  - ✓ Large amount of data
  - ✓ Real time data
  - ✓ Tool to analyse and interpret data
- **Tools/solution:**
  - ✓ Remote Sensing
  - ✓ GIS



Solution....??

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Coronavirus and space technologies: Satellites to fight with COVID-19!

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News / India / Chennai: Institute of remote sensing helps state govt to map Covid-19 cases

Chennai: Institute of remote sensing helps state govt to map Covid-19 cases

Experts are using this data to prepare a grid that will help the state agencies work on th containment zones.

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Monitoring the COVID-19 Crisis From Space

A comparison of nighttime light emissions across China during the early months of COVID-19 shows how satellite sensing data can provide early detection of trouble.

NASA to study impact of Covid-19 on environment using satellite imagery, hopes to map user behaviour during pandemic

The results of these studies could play a major role in forming public policy.

By *Rohan Abraham*, ET Bureau | Last Updated: May 15, 2020, 06.06 PM IST

Save

Solution...??



Huoshenshan Hospital in Wuhan, China was built in 10 days.
Source: GF-2 | Resolution: 0.8m



Leishenshan Hospital in Wuhan, China was built in 14 days.
Source: GF-2 | Resolution: 0.8m

Image Source/Reference/Courtesy: SpaceWILL (Google)

Solution...??

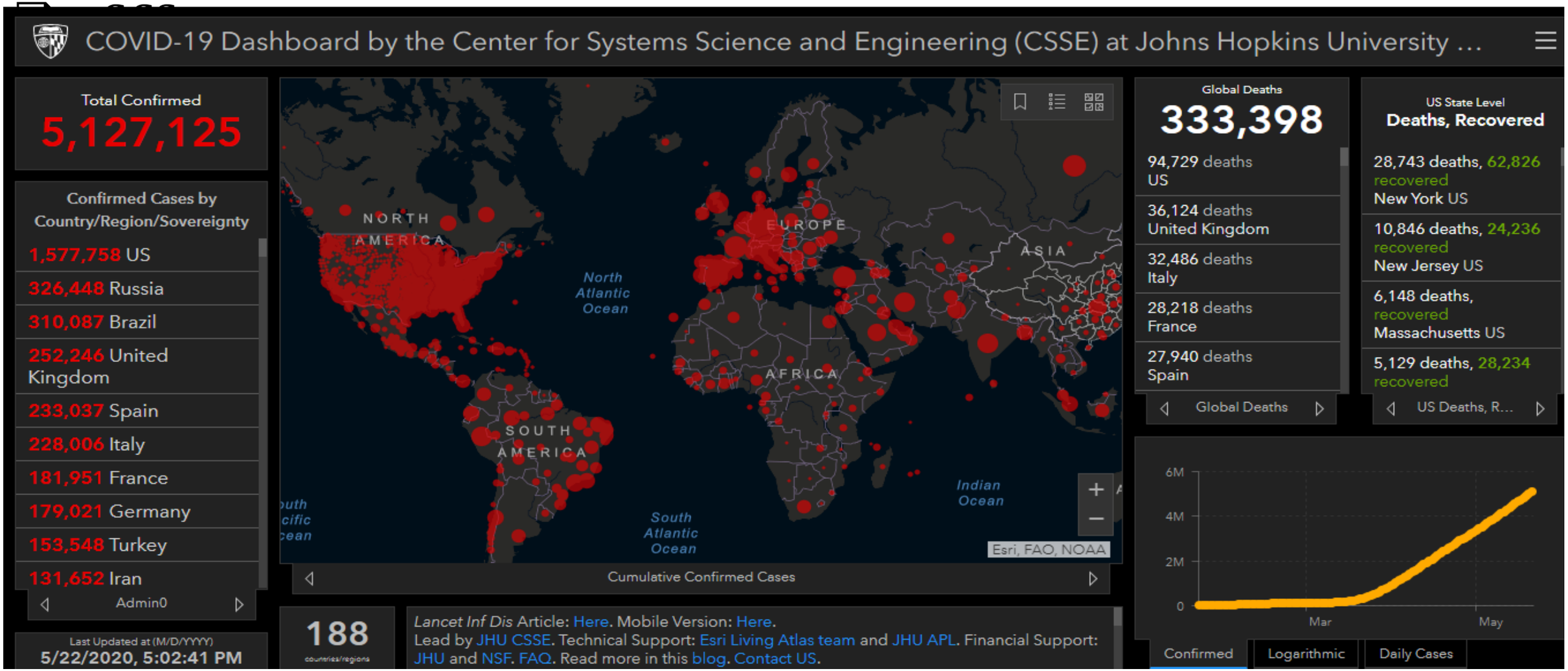


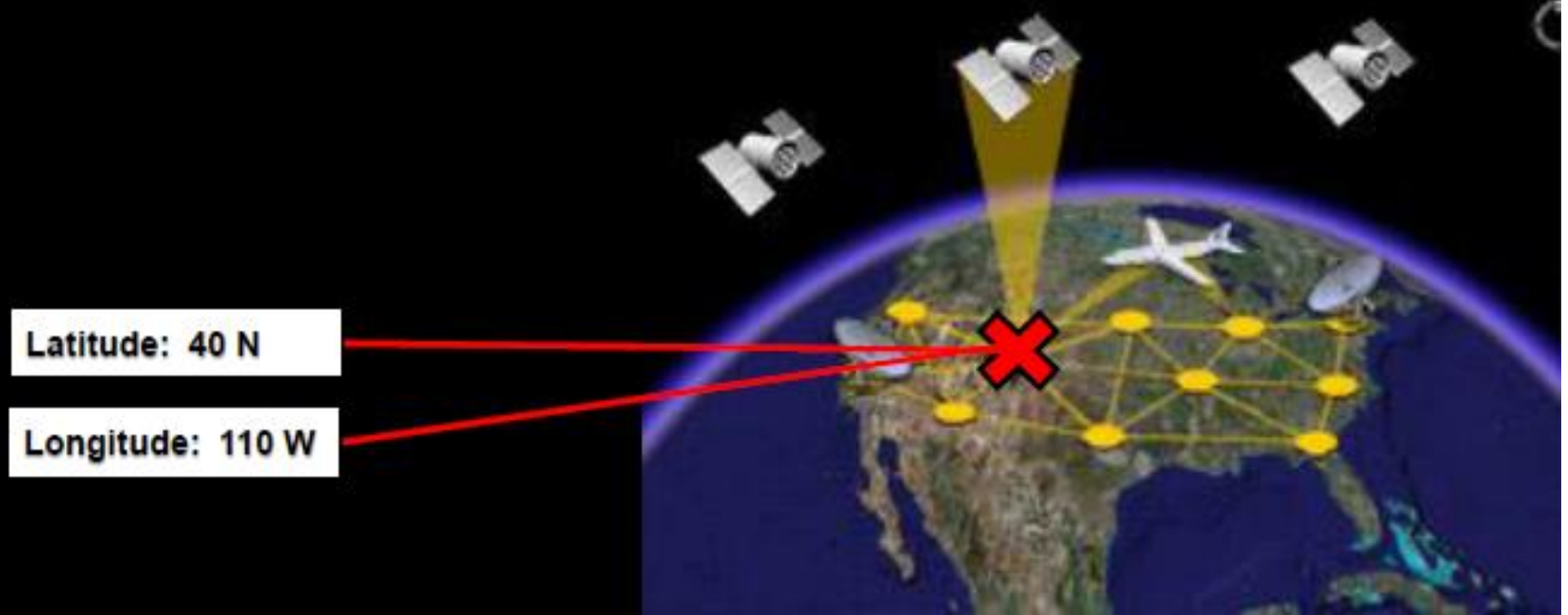
Image Source/Reference/Courtesy: ESRI

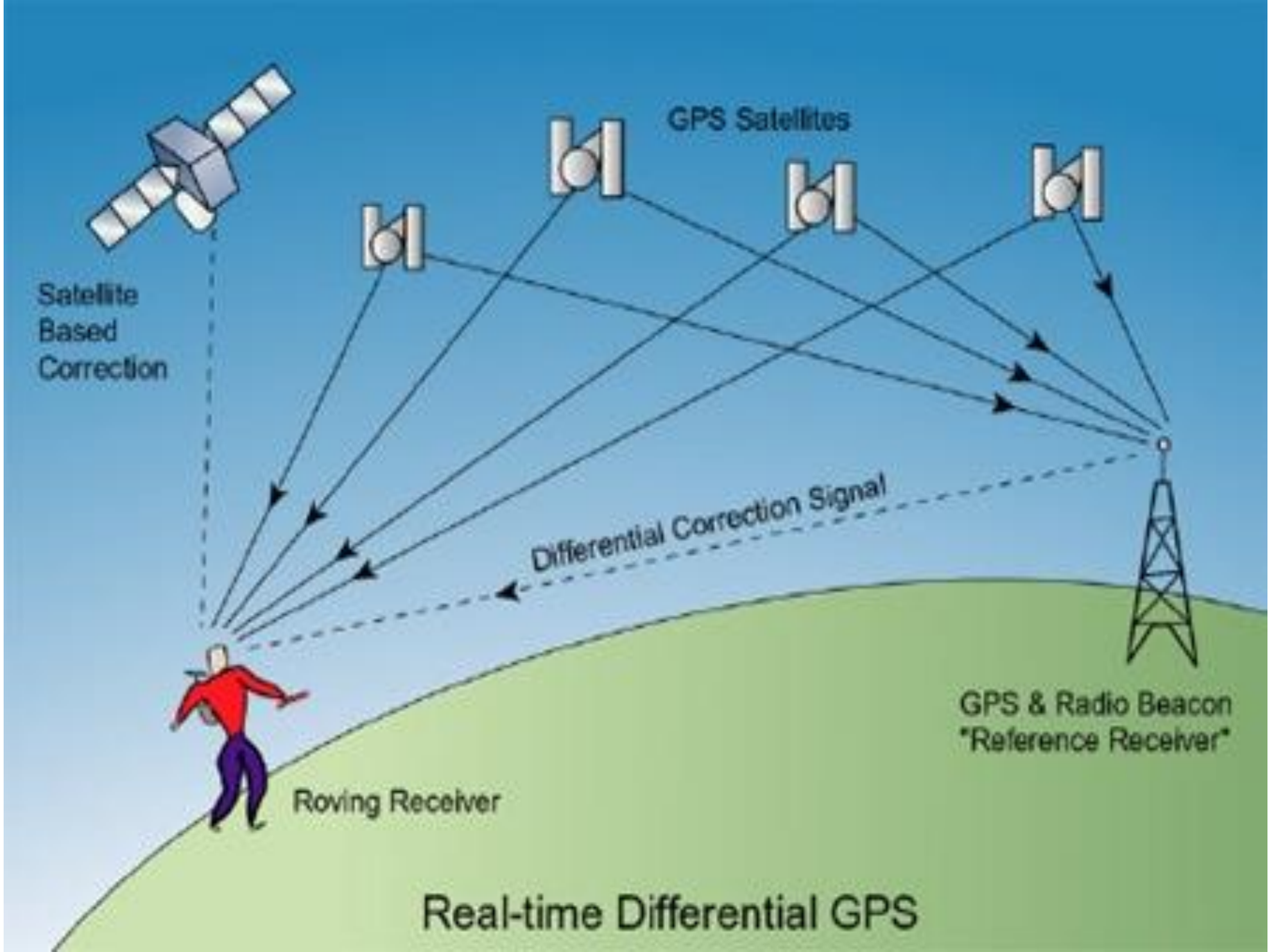
GPS – Where am I?

Global

Positioning

System




GPS Satellites
Satellite
Based
Correction
Differential Correction Signal
GPS & Radio Beacon
"Reference Receiver"
Roving Receiver
Real-time Differential GPS

Who uses GPS?





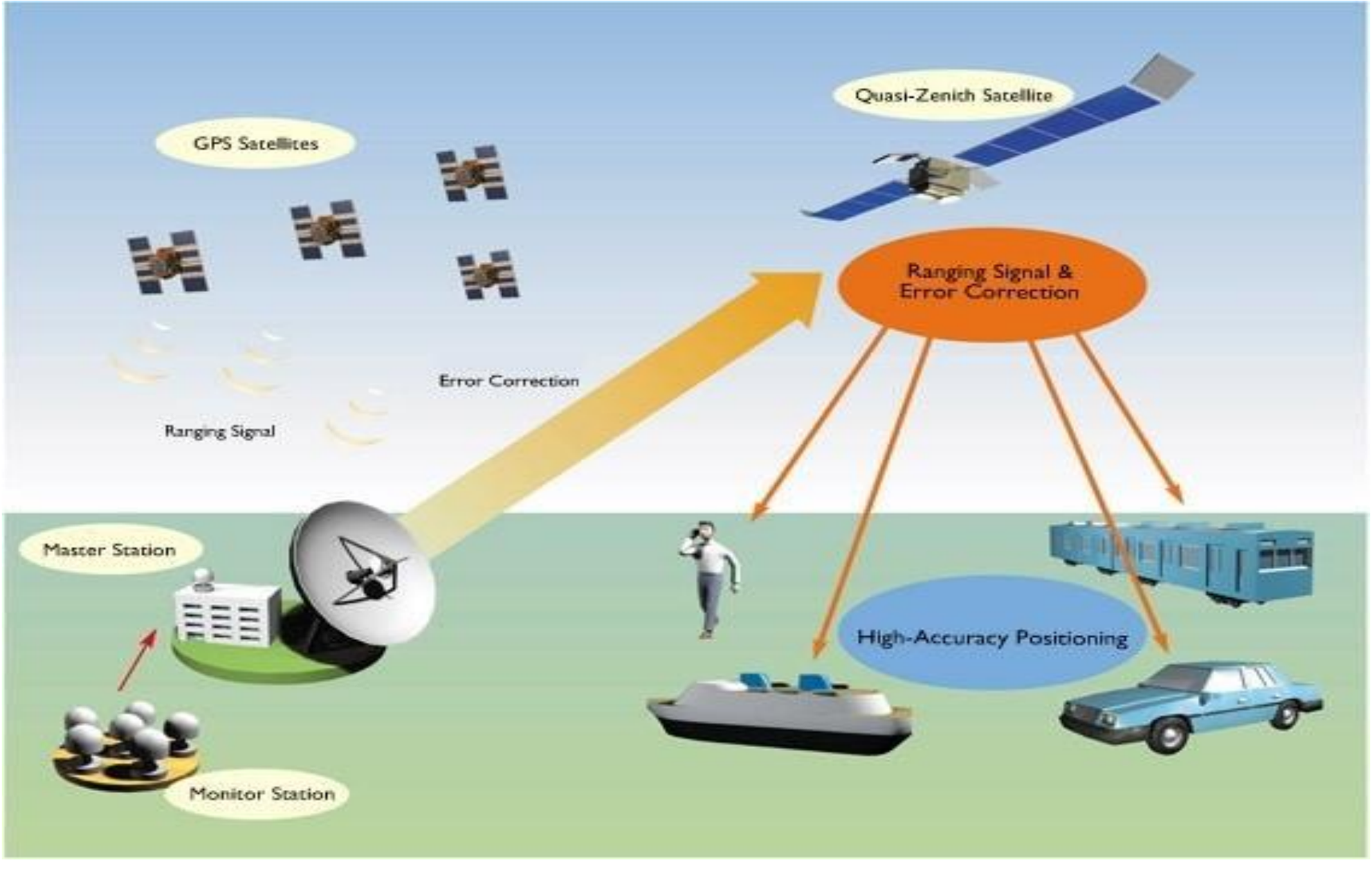

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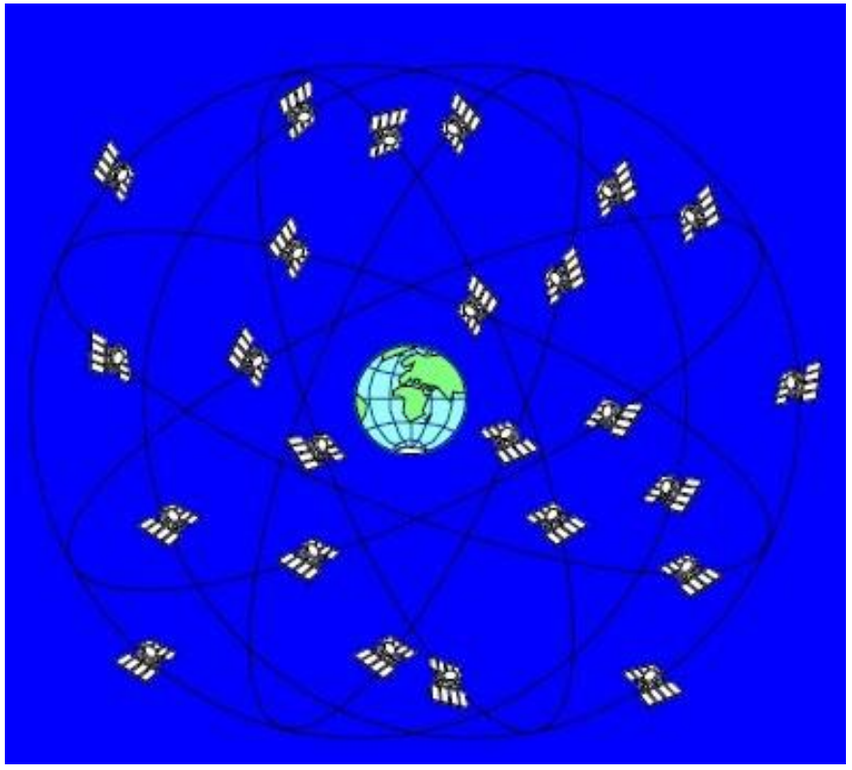


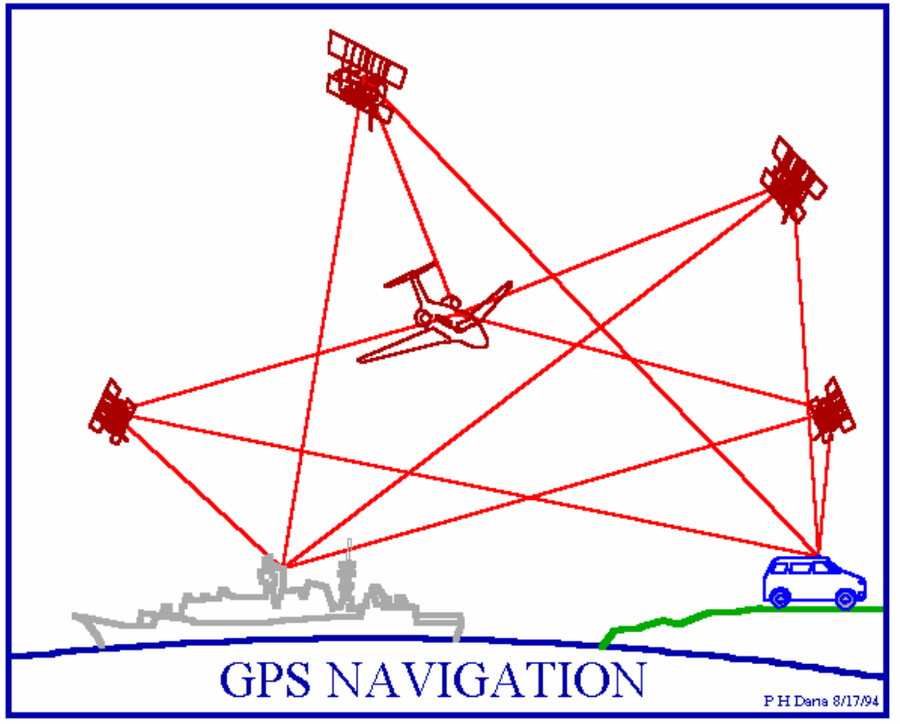
Global Navigation Systems



Global Navigation Systems:

- The **Global Positioning System (GPS)** is a space- based satellite navigation system that provides location and time information in all weather conditions, anywhere on or near the Earth where there is an unobstructed line of sight to four or more GPS satellit





Global Positioning satellite System (GPS)

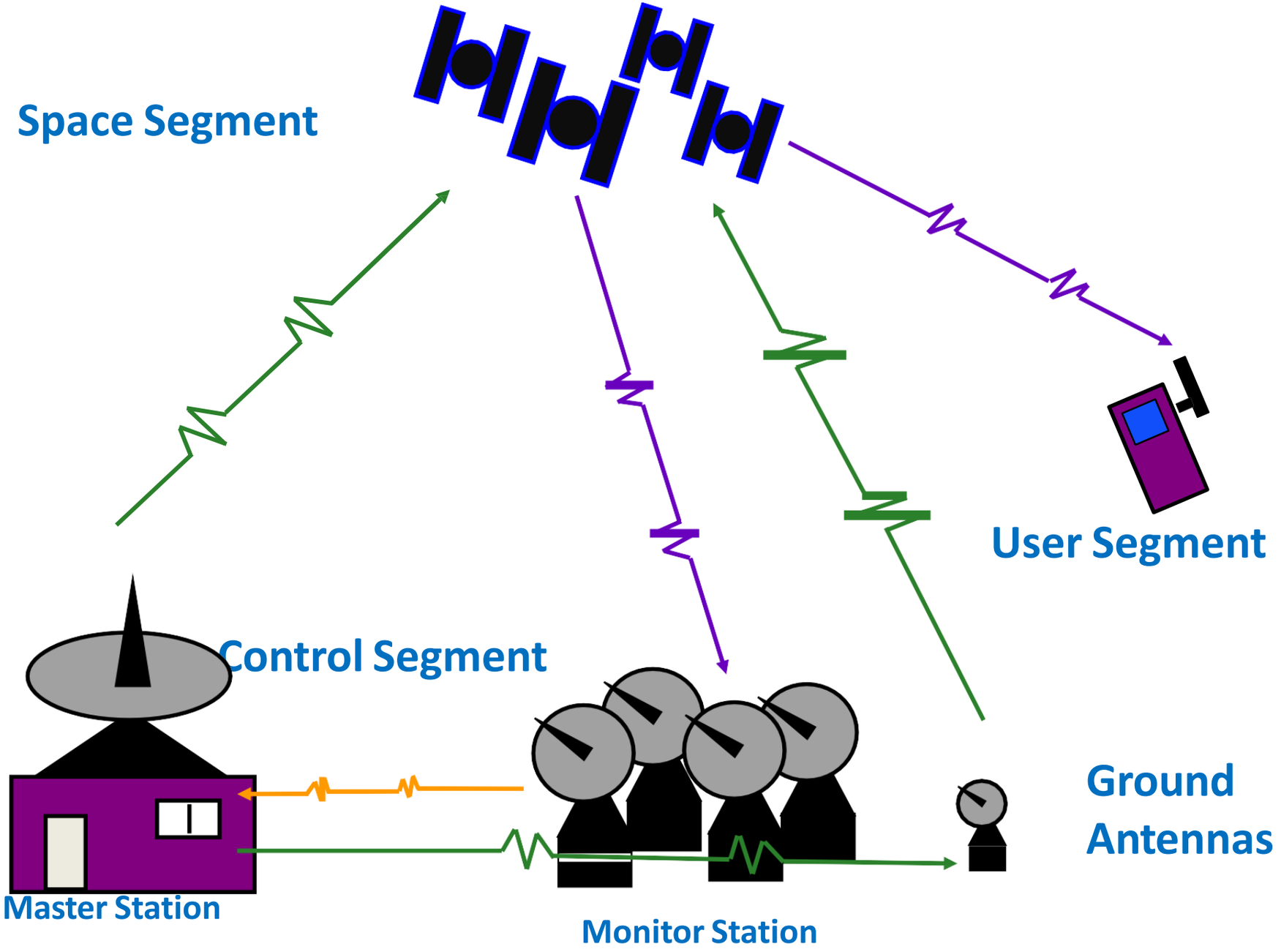
GPS

- Stands for Global Positioning System
- GPS is used to get an exact location on or above the surface of the earth (1cm to 100m accuracy).
- Developed by DoD (Department of Defense, U.S.) and made available to public in 1983.
- GPS is a very important data input source.
- GPS is one of two (soon to be more) GNSS – Global Navigation Satellite System

GNSS

- NAVSTAR – U.S. DoD (“GPS”)
- GLONASS – Russian system
- Galileo – European system (online in 2019?)
- Compass/BeiDou-2 – Chinese system in development (operational with 10 satellites as of December, 2011; 35 planned)
- GPS and GLONASS are free to use!

Segments of GPS



Data Models

- Raster Model
  - The first GIS model developed
  - Based on grids of cells that are assigned values and grouped into layers
- Vector Model
  - Uses points, lines, and polygons define data classes
  - Grouped into themes

GNSS Comparison

- GLONASS
  - 24 satellites (100% deployed)
  - 3 orbital planes
- GPS
  - 31 satellites (>100% deployed)
  - 6 orbital planes

System Components

- Receiver
  - Receives satellite signals
  - Compiles location info, ephemeris info, clock calibration, constellation configuration (PDOP)
  - Calculates position, velocity, heading, etc…
- Data Collector
  - Stores positions (x,y,z,t)
  - Attribute data tagged to position
- Software
  - Facilitates file transfer to PC and back
  - Performs differential correction (post-processing)
  - Displays data and

Thank you for watching...!!!

Any Questions..???

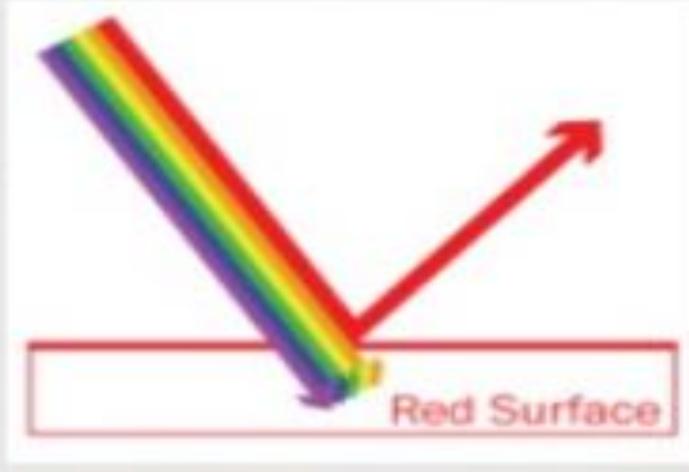


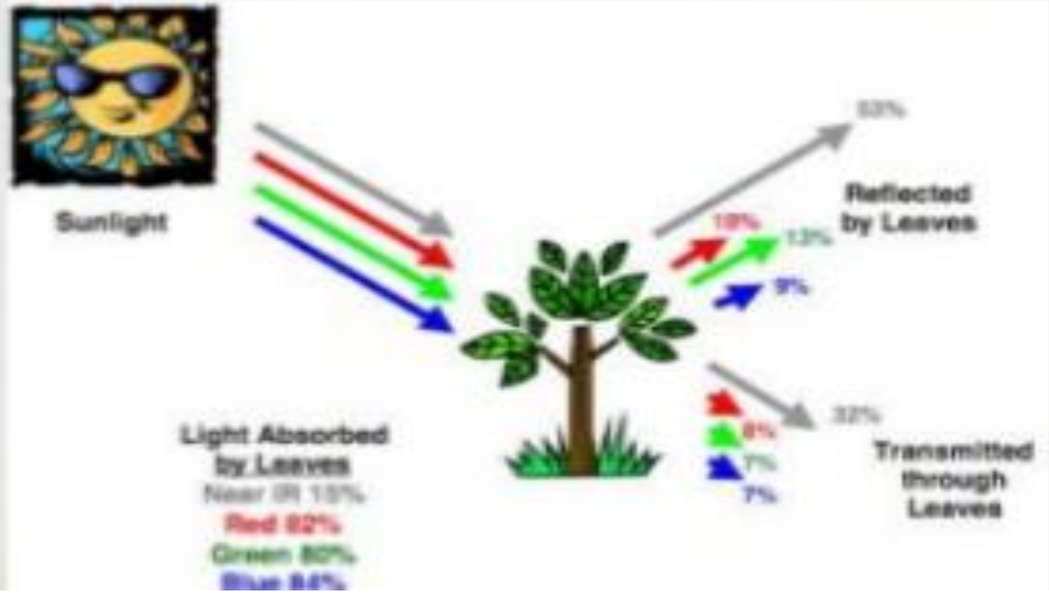
Principle behind RS

Electromagnetic energy reaching the earth's surface from the Sun is **reflected**, transmitted or absorbed.

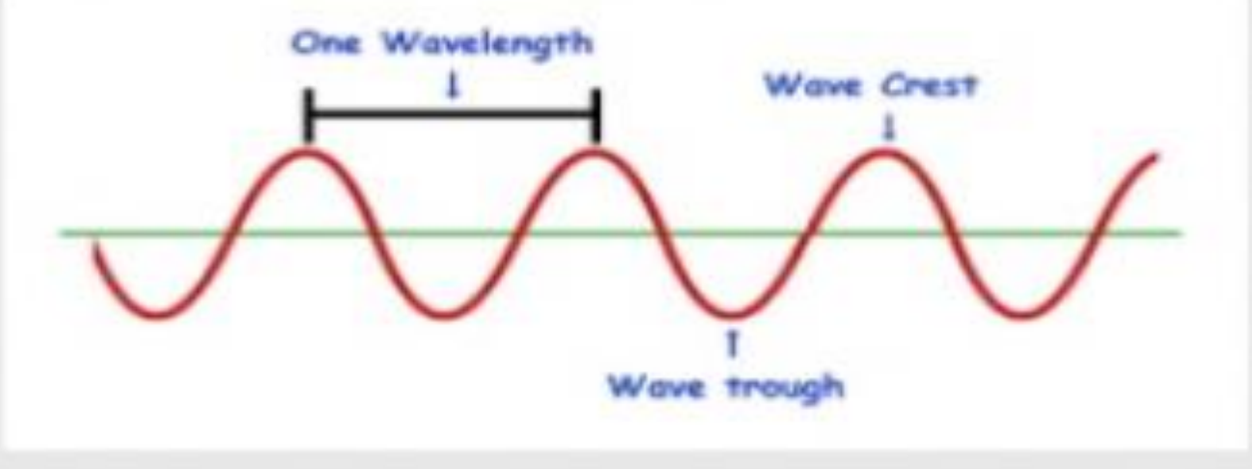
Specific targets have an individual and characteristic manner of interacting with incident radiation that is described by the spectral response of that target.

Egs-soils of differed types, water with varying degrees of impurities, or vegetation of various species



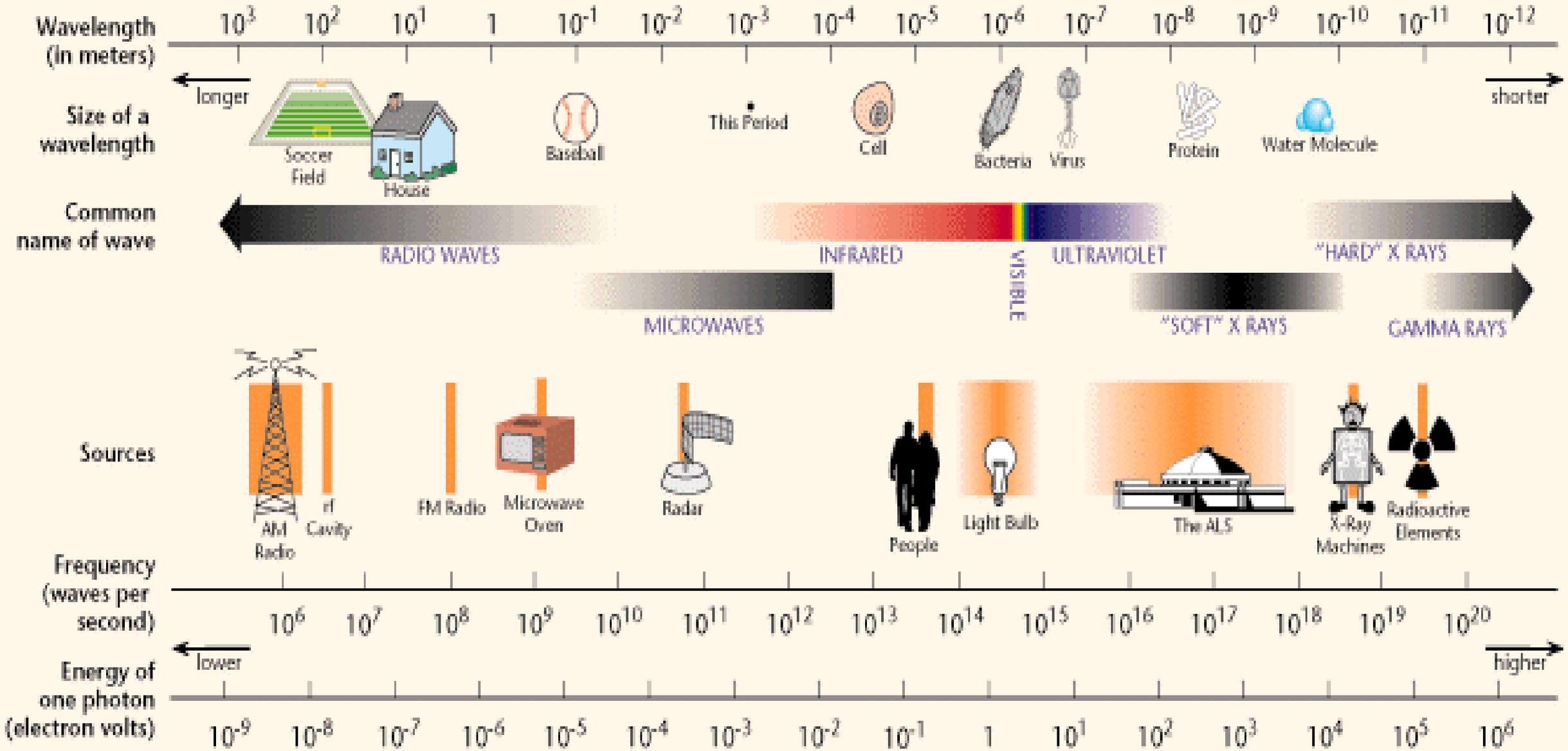


- Electromagnetic Radiation (EMR) like radio waves, infrared (heat) waves make characteristic patterns as they travel through space. Each wave has a certain shape and length. The distance between peaks (high points) is called wavelength.



- The light which our eyes - our "remote sensors" - can detect is part of the **visible spectrum**. The visible wavelengths cover a range from approximately 0.4 to 0.7 µm (micrometre; 1 µm= 1×10^{-6} of a metre).

THE ELECTROMAGNETIC SPECTRUM



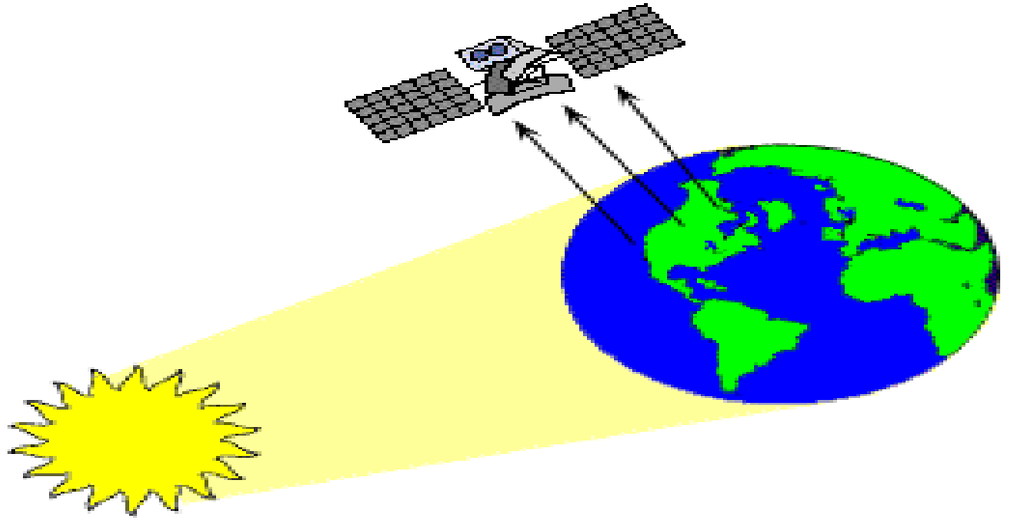
- The sun is a source of energy or radiation, which provides a very convenient source of energy for remote sensing.
- The sun energy is either reflected, as it is for visible wavelengths, or absorbed and then re-emitted, as it is for thermal infrared wavelengths.

Types of remote sensing

1. Passive remote sensing
2. Active remote sensing.

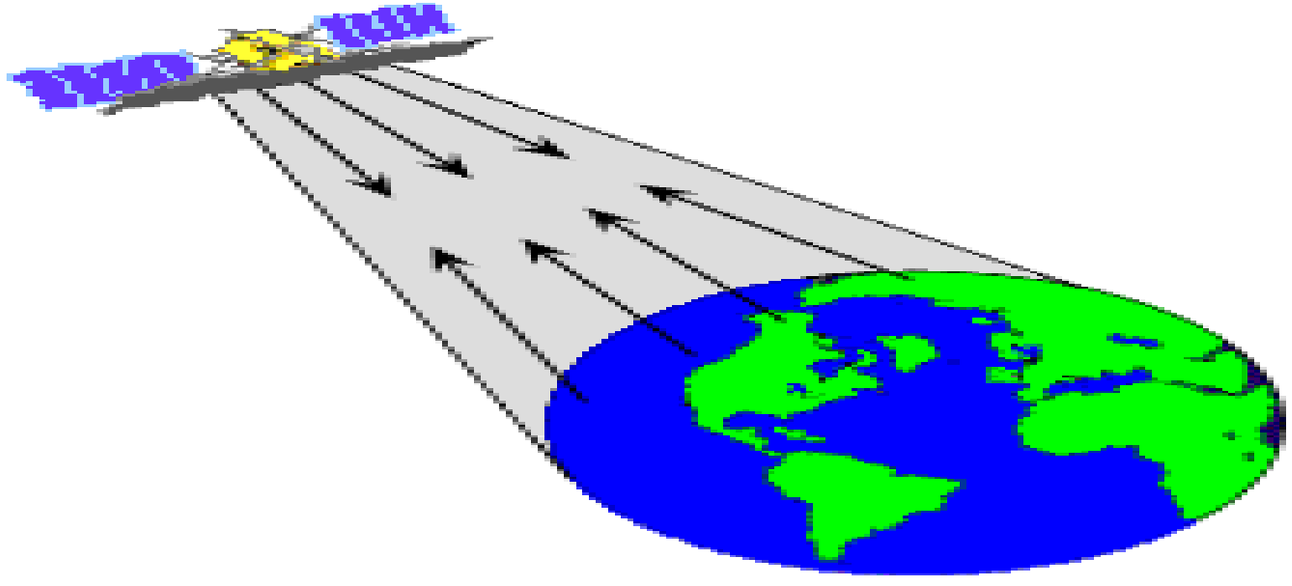
1. Passive Remote Sensing

- Passive sensors detect natural radiation that is emitted or reflected by the object or surrounding area being observed.
- Reflected sunlight is the most common source of radiation measured by passive sensors.
- Examples of passive remote sensors include film photography, infrared, and radiometers.



2. Active Remote Sensing

- Emits energy in order to scan objects and areas where upon a sensor then detects and measures the radiation that is reflected from the target.
- RADAR is an example of active remote sensing where the time delay between emission and return is measured, establishing the location, height, speeds and direction of an object



What kinds of devices collect the data?

- Aircraft
  - High altitude
  - Low altitude
- Spacecraft:
  - Landsat
  - SPOT
  - Weather satellites
  - GeoEye-1

Geosynchronous Orbit

when the satellite moves at the same speed as the spinning earth – results in the camera staying over the same spot of the earth

Who uses Remote Sensing and why?

- the geographer, who looks for changes on the Earth's surface that need to be mapped;
- the forester, who needs information about what type of trees are growing and if they have been affected by disease or fire;
- the environmentalist, who wants to detect, identify and follow the movement of pollutants such as oil slicks on the ocean;
- the geologist, who is interested in finding valuable minerals;

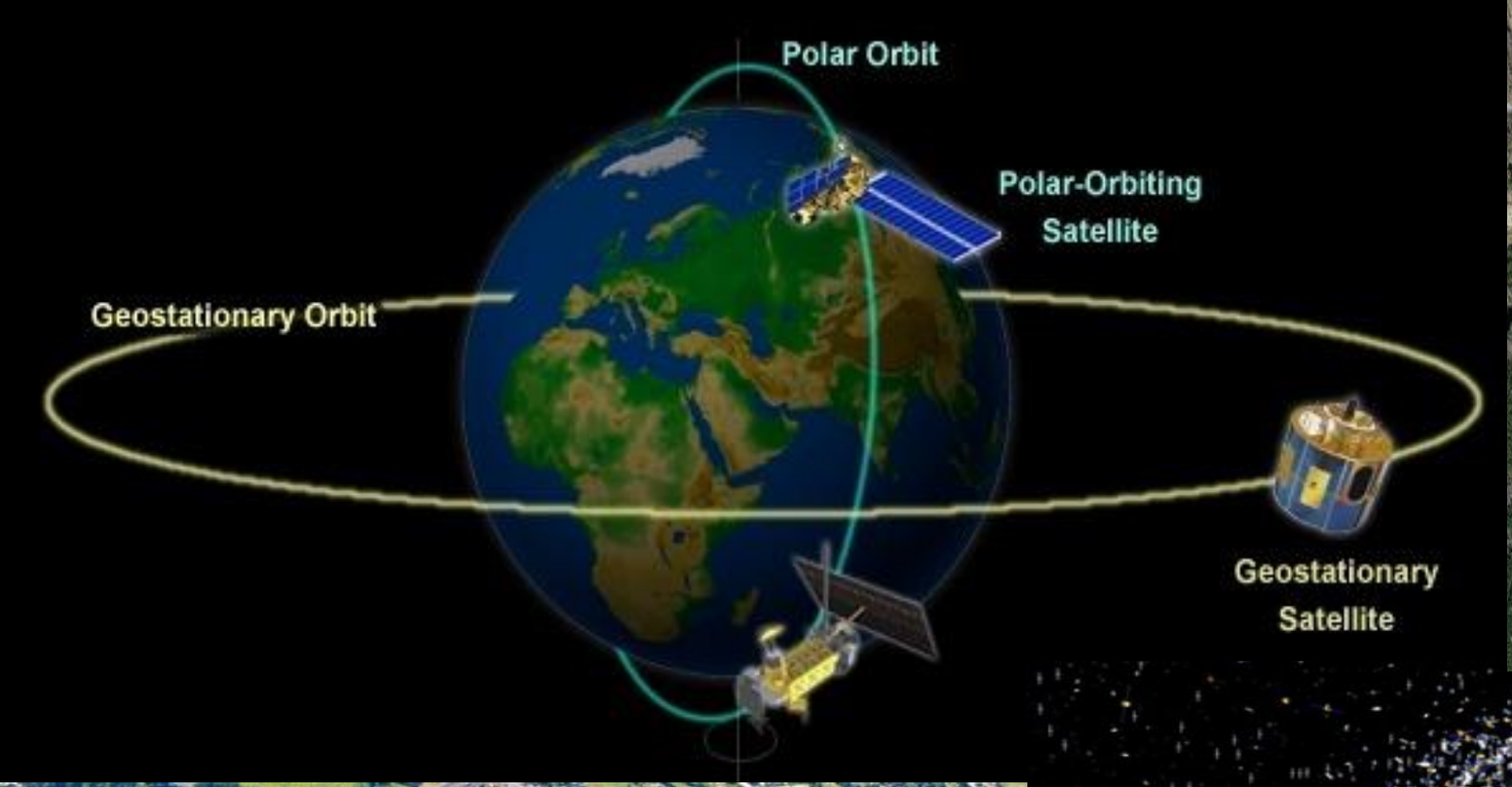
- the farmer, who wants to keep an eye on how his crops are growing and if they've been affected by drought, floods, disease or pests;
- the ship captain, who needs to find the best route through the northern ice packs;
- the firefighter, who sends out his crews based on information about the size and movement of a forest fire.
- the Urban Planner, who wants to map and monitor land cover, land use, morphology (the study of the form of human settlements and the process of their formation and transformation) etc.

Platforms

- Platforms are used to house the sensors which obtain data for remote sensing purposes.
- The distance between the target being imaged and the platform, plays a large role in determining the detail of information obtained and the total area imaged by the sensor.

- Platforms are-
  - Ground based
  - Airborne eg. Aircraft, Drone
  - Space borne eg. Satellite




Polar Orbit
Polar-Orbiting Satellite
Geostationary Orbit
Geostationary Satellite


Before
After





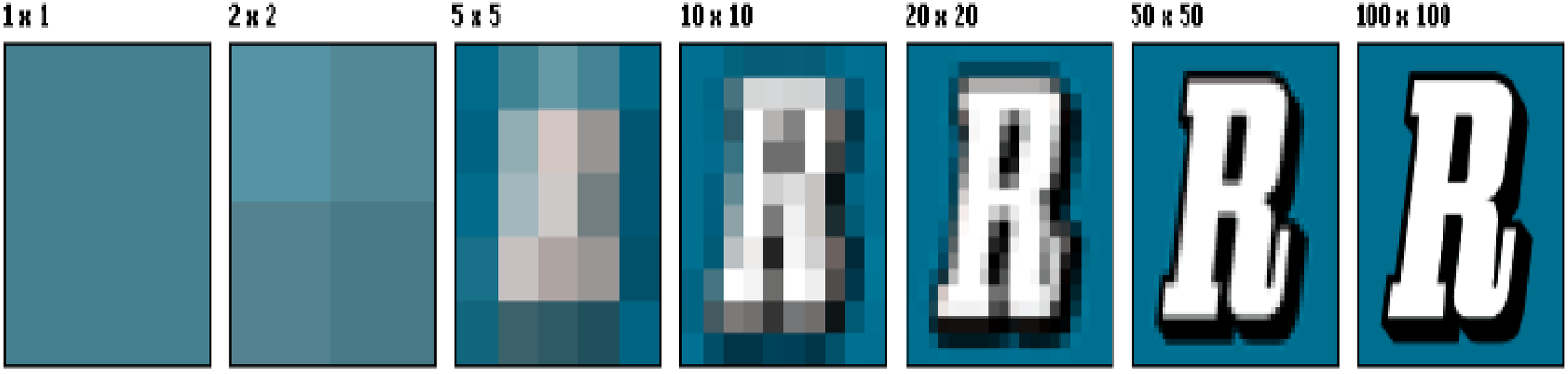
Sl. No.	Field of application	Useful interpreted information	Helpful in
1.	Irrigation Engineering	Crop area, Crop yield, Crop growth condition, Crop areas that are water stressed and are in need of water	Estimating the amount of irrigation water that is to be supplied to an irrigated area over different seasons
2.	Hydrology	Different types of soils, rocks, forest and vegetation of a water shed, soil moisture	Estimating runoff from a watershed, where the land-cover type and soil moisture would decide the amount that would infiltrate
3.	Reservoir sedimentation	Plan views of reservoir extent at different times of the year and over several years	Estimating the extent of sedimentation of a reservoir by comparing the extent of reservoir surface areas for different storage heights
4.	Flood monitoring	Flood inundated areas	Flood plain mapping and zoning
5.	Water Resources Project Planning	Identification of wasteland (from MSS images), mapping of infrastructure features (from PAN images) like existing roads, embankments, canals, etc. apart from plan view of a river	Recent information helpful in planning and designing of a water resources project based on the present conditions of the project area

Image resolution

- It is an umbrella term that describes the detail an image holds.
- The term applies to raster digital images, film images, and other types of images.
- Higher resolution means more image details.
- The resolution of digital images can be described in many different ways.
- The term **resolution is often used for a pixel count in digital imaging, even though American, Japanese, &** international standards specify that it should not be so used, at least in the digital camera field.

Effect of pixel resolutions

Below is an illustration of how the same image might appear at different pixel resolutions, if the pixels were poorly rendered as sharp squares (normally, a smooth image reconstruction from pixels would be preferred, but for illustration of pixels, the sharp squares make the point better).




Spatial Resolution
33km 25km 17km 9km 1km


Spatial Resolution
33km 25km 17km 9km 1km


Spatial Resolution
33km 25km 17km 9km 1km

Image Interpretation

Image Interpretation

- We must be able to extract meaningful information from the imagery.
- Interpretation involves the identification and/or measurement of various targets in an image
- Targets in remote sensing images may be any feature or object which can be observed in an image
- The target must be distinguishable.

Interpretation strategies

- Field observation.
  - Required when the image and its relationship to ground conditions are imperfectly understood
- Direct recognition.
  - Interpreter derives information directly from inspection of the image
- Inference
  - Based on visible features can derive information about invisible features
  - Certain relationships and identify features

Elements of imageinterpretation

- Image analysis requires explicit recognition of eight ***elements of image interpretation*** that form the framework and understanding of an image
  - Shape
  - Size
  - Tone
  - Texture
  - Shadow
  - Site
  - Association
  - pattern

Shape

- The outline of a feature
  - Important to note that shape depends on
  - perspective
  - Overhead perspective, introduces scale effect





Size

- The first to the dimensions of a feature
- ***Relative size*** determined by comparing the object with familiar nearby features
- ***Absolute size*** refers to the use of the aerial image to derive measurements



Tone

- Refers to the average brightness of an area or, in the case of color imagery, to the dominant color of the region
  - Smooth surfaces behave like ***specular reflectors***, they tend to reflect radiation in a single direction
  - These features may appear bright or dark
  - Rough surfaces behave this ***diffuse reflectors***.



Texture

- Refers to the variation in tone over a surface or the apparent roughness of the surface as seen in the photo
- Created by micro shadows in small irregularities in the surface.





Shadow

- Refers to large distinctive shadows that revealed the outline of a future as projected onto a flat surface.
  - Depends on the nature of the object, angle of illumination, perspective, and slope of the ground surface











Site

- Refers to a futures position with respect to topography and drainage.
  - Some things occupy a distinctive topographic position because of their function
    - Sewage treatment facilities at the lowest feasible topographic position.
    - Power plants located adjacent to water for cooling









Association

- Association refers to the distinctive spatial interrelationships between features
  - Schools often associated with athletic fields.
  - Large parking lots often associated with malls







Pattern

- Refers to distinctive arrangement of features
  - Orchards have trees plant can rows
  - Mobile home parks have rectangular buildings arranged in rows

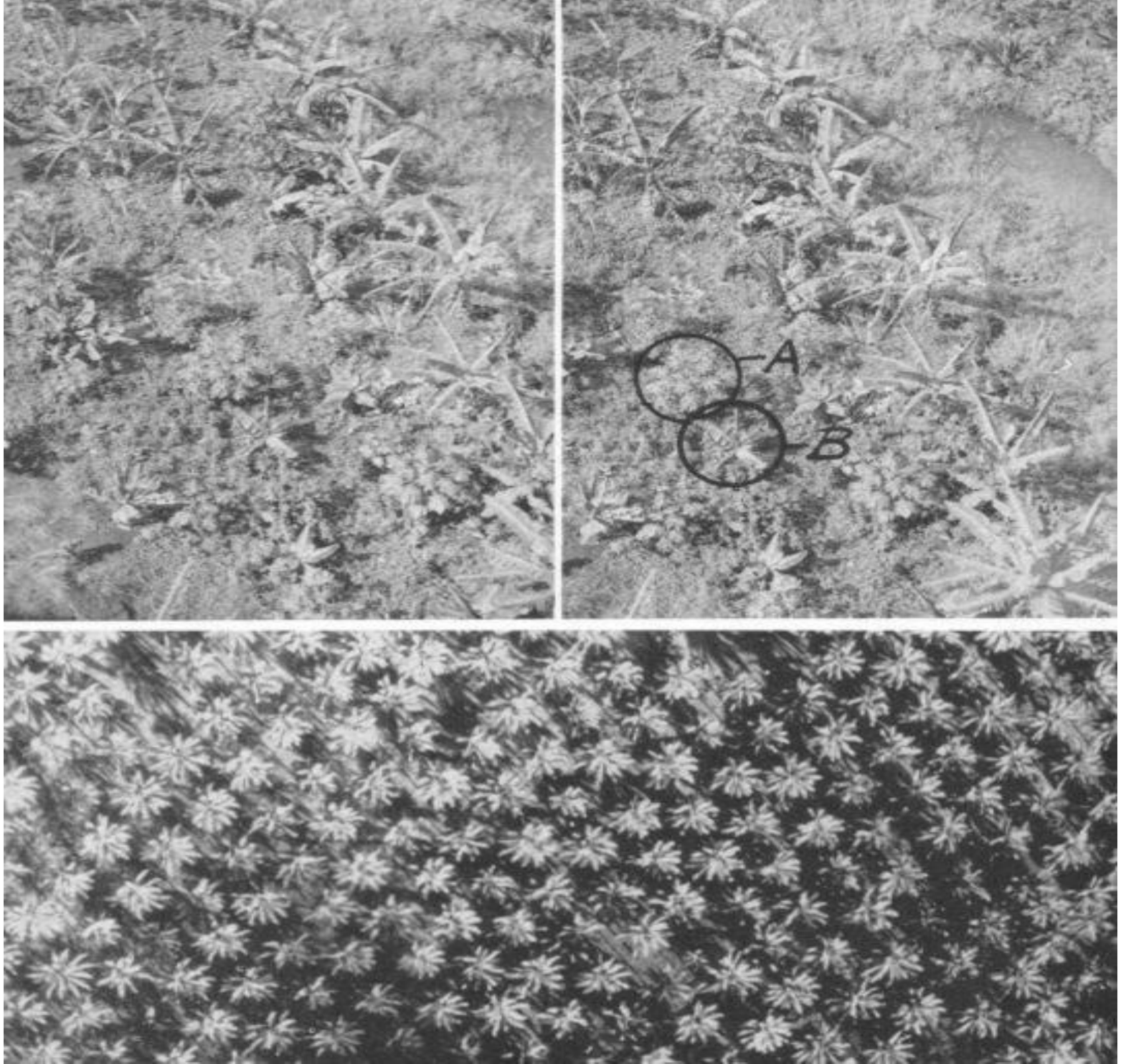











A
B

Application of GIS in Hydrology

I. **Surface Water Hydrology**

➢ ***Data Required :***

- ***Terrain***
  - Digital elevation models
  - Slope and aspect
  - Watersheds and sub-catchments
  - Drainage network
- ***Soil***
  - Permeability
  - Layer depth
  - Soil textural and Soil water content

- ***Precipitation & climate***
  - Rain-gauge data
  - Gauge locations & context
  - Statistics(e.g., *intensity, duration)*
  - Temperature
  - Evaporation & transpiration

Analysis of Data & Final Output :

- Watershed Management
  - Terrain modeling
  - Flow modeling
  - Debris flow probability

- Flood Management
  - Flood plain delineation
  - Channel characteristics
  - Inundation modeling
  - Infrastructure analysis
  - Risk modeling and mitigation

II. Groundwater Modeling

- Data Required :
  - Weather
  - Topography
  - Land cover type
  - Water levels and geologic data from no. of groundwater wells
- Analysis & Output
  - Ground water model
  - Groundwater development and
  - Artificial recharge sites

Example

- **Project Planning for a Storage Structure**
  - In this example, a dam is proposed to be constructed across a river, for which the following information may be required:

    i. Watershed area contributing to the project site
    ii. Reservoir surface area and volume, given the height of the dam
    iii. Villages that may be inundated under reservoir

- For the above, the following themes may be stored in a GIS:
  - Elevation contours of the watershed area, including the project site
    - *For this, DEM is required*
  - Satellite image derived land-use map of the watershed
    - *For this, Satellite images which shows land use/land cover data is required*
  - Village boundary map, showing location of habitation clusters
    - *For this, digital map which shows village boundaries and other important structure is required*

- Using the above data, one may obtain desired in information as follows:
  i. Watershed area may be found by using the elevation contour data, and using a suitable GIS software that has a tool to delineate the watershed boundary.
  ii. Once the boundary is identified, the area calculation tool may be used in the GIS software to calculate the watershed area.
  iii. Reservoir surface area can similarly found using the area calculation tool.

iv. Volume calculation tool of the GIS software may be used to find out the storage volume, which is the space between a plane at the reservoir surface and the reservoir bottom.

v. By overlying the reservoir extent over the village boundary map and the locations of habitation clusters one may identify the villages that are likely to be inundated once the reservoir comes up.

vi. The area of the cultivable village farms that would be submerged may also be similarly identified, as it would be required to pay compensation for the loss to the villagers.

Any Questions……??